

Lecture 5

Microeconomic Theory III – ECON 709

Fall 2026

Recap' Lecture 4

3 equivalent definitions of continuity of a preference relation:

- ▶ A rational preference relation $\succsim$ on X is continuous if whenever $a \succ b$ (namely, it is not true that $b \succsim a$), there are balls B_a and B_b around a and b , respectively, such that for all $x \in B_a$ and $y \in B_b$, $x \succ y$.
- ▶ A rational preference relation $\succsim$ on X is continuous if it is preserved under limits. That is, for any sequence of pairs $\{(a_n, b_n)\}_{n=1}^{\infty}$ with $a_n \succsim b_n$ for all n , $a_n \rightarrow a$ and $b_n \rightarrow b$, we have $a \succsim b$.
- ▶ A rational preference relation $\succsim$ is continuous if, for all $x \in X$, the upper contour set $\{y \in X : y \succsim x\}$ and the lower contour set $\{y \in X : x \succsim y\}$ are closed.

Debreu's Theorem (1954)

Debreu shows that continuity of a rational preference relation is sufficient for the existence of a utility representation.

Proposition 8

Assume X is a convex subset of $\mathbb{R}^n$. Let $\succsim$ be a continuous and rational preference relation on X . Then there is a continuous utility function $u(x)$ which represents $\succsim$.

- ▶ classical result in economic theory
- ▶ we will prove the result with the following additional simplifying assumptions:
 - ▶ $X = \mathbb{R}_+^K = \{x = (x_1, \dots, x_K) \mid \text{for all } k, x_k \geq 0\}$
 - ▶ $\succsim$ is monotonic, i.e. for all x and all $y \in X$, if $x_k \geq y_k$ for all k , then $x \succsim y$, and if $x_k > y_k$ for all k , then $x \succ y$.
"More is better", a common assumption in economics, discussed in more detail later
- ▶ for a proof in the more general setting, see Debreu (1954, 1960)

Proof of Proposition 8 with $X = \mathbb{R}_+^K$ and monotonicity

Proof strategy:

- ▶ construct one function that is continuous and that represents the given preferences
- ▶ first construct a function $u : X \rightarrow \mathbb{R}$, second show that u represents the preferences, third show that u is continuous
- ▶ How do we construct the utility representation here?
For every $x \in X$, we will find one unique indifferent bundle on the main diagonal.
Then we assign to each bundle the amount in its indifferent bundle on the main diagonal as utility value.
(That is, we map all bundles in an order-preserving way onto the main diagonal line and assign each bundle its main diagonal value as utility value.)

First, construct function $u : X \rightarrow \mathbb{R}$.

1. Show that for every $x \in X$, there is $m^* \in X$ on the main diagonal (equal quantities of all commodities), s.t. $x \sim m^*$ (mapping all bundles onto the diagonal):
 - ▶ by monotonicity, $x \succsim 0 = (0, \dots, 0)$ and $M = (\max_k \{x_k\}, \dots, \max_k \{x_k\}) \succsim x$
 - ▶ both 0 and M are on the main diagonal

- construct inductively the sequence $\{(y^n, z^n, m^n)\}$ of elements in X as follows:

Start with $y^0 = M$, $z^0 = 0$ and the midpoint
 $m^0 = 1/2y^0 + 1/2z^0$.

If $m^n \sim x$, then we found m^* .

If $m^n \succ x$, let $y^{n+1} = m^n$ and $z^{n+1} = z^n$.

If $x \succ m^n$, let $y^{n+1} = y^n$ and $z^{n+1} = m^n$.

(One of the above must be true because of completeness.)

In any case define $m^{n+1} = 1/2y^{n+1} + 1/2z^{n+1}$.

If none of the points $m^n \sim x$, then $y^n \succ x \succ z^n$ for all n .

Both sequences (y^n) and (z^n) converge to some m^* (the same!) on the main diagonal between 0 and M

because $z^n \leq z^{n+1}$ and $y^{n+1} \leq y^n$ with an inequality for one of them, and $z^{n+1} < y^{n+1}$ for all n

(the distance between y^n and z^n is converging to zero).

Since $\succsim$ is continuous, we know that $\{(y^n, x^n)\}$ with $x^n = x$ and $y^n \succsim x$ for all n , $y^n \rightarrow m^*$ implies $m^* \succsim x$ and $\{(x^n, z^n)\}$ with $x^n = x$ and $x \succsim z^n$ for all n , $z^n \rightarrow m^*$ implies $x \succsim m^*$.
Thus $m^* \sim x$.

2. For every x there is a unique m^* on the main diagonal such that $m^* \sim x$ (we check that we mapped each bundle to exactly one point on the main diagonal):
 - Let $m^* \sim x$. For every $m \ll m^*$ (m has every element strictly smaller than m^*) on the main diagonal, $m^* \succ m$ by monotonicity and hence $x \succ m$ by transitivity. Similarly for $m \gg m^*$.
3. Denote the unique m^* on the diagonal such that $m^* \sim x$ by $(t(x), \dots, t(x))$. Let $u(x) = t(x)$. (assigning the utility value to x)

Second, verify that the function $u : X \rightarrow \mathbb{R}$ is a utility representation of $\succsim$.

- ▶ $x \succsim y$
 $\Leftrightarrow (t(x), \dots, t(x)) \succsim (t(y), \dots, t(y))$ (by transitivity)
 $\Leftrightarrow t(x) \geq t(y)$ (by monotonicity)
 $\Leftrightarrow u(x) \geq u(y)$

Third, prove that u is continuous.

- ▶ Reminder: A function $u : X \rightarrow \mathbb{R}$ is continuous if for all x and any sequence (x^n) with $x^n \rightarrow x$, we have $u(x^n) \rightarrow u(x)$.
- ▶ Let (x^n) be a sequence converging to x and assume by contradiction that $t(x^n)$ does not converge to $t(x)$.
- ▶ Since $x^n \rightarrow x$, there is a number M^* s.t. $(M^*, \dots, M^*) \succ x^n$ for all n .
- ▶ Then all members of the sequence $(t(x^n))$ are in the compact interval $[0, M^*]$, as $(t(x^n), \dots, t(x^n)) \sim x^n$.

- ▶ As $t(x^n) \not\rightarrow t(x)$ there is an $\epsilon > 0$ s.t. for every integer N there is $n > N$ such that $t(x^n)$ is outside the interval $(t(x) - \epsilon, t(x) + \epsilon)$ and thus there must be infinitely many t -values outside of that interval.
- ▶ Then, there is an infinite subsequence of (x^n) with all t -values in $[t(x) + \epsilon, M^*]$ (or all t -values in $[0, t(x) - \epsilon]$).
- ▶ This subsequence has an infinite subsequence (y^n) with $(t(y^n))$ converging to some $t' > t(x)$. (by Bolzano-Weierstrass-Theorem)
- ▶ Now, $y^n \rightarrow x$ (as (y^n) is a subsequence of (x^n) which converges to x) and $(t(y^n), \dots, t(y^n)) \rightarrow (t', \dots, t')$.
- ▶ Since $y^n \sim (t(y^n), \dots, t(y^n))$ for all n , $x \sim (t', \dots, t')$ by the continuity of preferences.
- ▶ However, $x \sim (t(x), \dots, t(x))$ and $t' > t(x)$, a contradiction. □

Any continuous utility function represents a continuous and rational $\succsim$

Proposition 9

If $\succsim$ on X is represented by a continuous utility function u , then $\succsim$ is rational and continuous.

Proof.

- ▶ $\succsim$ is rational as rationality is a necessary condition for a utility representation to exist
- ▶ $\succsim$ is also continuous (we show that C1 is satisfied):
 - ▶ Consider $a \succ b$. Then $u(a) > u(b)$.
 - ▶ Let $\epsilon = (u(a) - u(b))/2$. By the continuity of u , there is a $\delta > 0$ such that for all x distanced less than δ from a , $u(x) > u(a) - \epsilon$, and for all y distanced less than δ from b , $u(y) < u(b) + \epsilon$.
 - ▶ Thus, for all x and y within the balls of radius δ around a and b , respectively, $x \succ y$.



Having talked about tastes, attitudes, and opinions, we now turn to behavior...

Choice

based on: Rubinstein, lecture 3; MWG, sections 1.C and 1.D

Choice Problem and Context

- ▶ X is the grand set of possible alternatives
- ▶ A is a non-empty subset of X , which constitutes the *choice problem*: Which element of A is chosen?
- ▶ D is the collection of all possible choice problems $A \subseteq X$ which the decision maker could face, thus D is a set of nonempty subsets of X . D does not need to include all possible nonempty subsets of X , but only the relevant choice problems
- ▶ we will refer to a pair (X, D) as a *context*

Choice Problem and Context: Example

- ▶ suppose we are interested in a student's choice from the set of universities to which he has been admitted
- ▶ let $X = \{x_1, \dots, x_N\}$ be the set of all universities to which the student applied to
- ▶ a choice problem $A \subseteq X$ is interpreted as the set of universities to which he has been admitted
- ▶ if the fact that the student was admitted to some universities does not mean he will be admitted to others, then D contains the $2^N - 1$ nonempty subsets of X
- ▶ if, for example, the universities are listed according to difficulty of getting into (x_1 being the most difficult) and if the fact that the student is admitted to x_k means that he is admitted to all less "prestigious" universities, that is, to all x_l with $l > k$, then D consists of the N sets $A_1, \dots, A_N$ where $A_k = \{x_k, \dots, x_N\}$.

Choice Rule

- ▶ given a context (X, D) , a choice rule is a correspondence C which assigns to every $A \in D$ a nonempty subset of chosen elements $C(A) \subseteq A$, that is $\emptyset \neq C(A) \subseteq A$
- ▶ $C(A)$ is the non-empty set of acceptable choices for the decisionmaker when he is presented with choice problem A
 - ▶ If $C(A)$ is a singleton, then this single element is the DM's choice. Back to our example: If $C(\{McGill, Toronto, Vancouver\}) = \{McGill\}$, then McGill is your choice when admitted to $\{McGill, Toronto, Vancouver\}$.
 - ▶ If $C(A)$ contains more than one element, then this does not mean that the DM chooses all of the elements in $C(A)$ at once, but rather each element is an equally acceptable good choice for the DM when presented with A . No unique choice prediction. "Equilibrium" interpretation: If some $x \in C(A)$ is chosen, then the decision maker has no desire to move away from his choice; he would, however, always swap an $x \notin C(A)$ for an $x \in C(A)$. Back to our example: If $C(\{McGill, Toronto, Vancouver\}) = \{McGill, Toronto\}$, then Toronto and McGill are equally acceptable choices for you.

Rational Choice

- ▶ In economics we usually think of an agent's decision process as a *rational choice* which goes through the following stages:
 1. the agent has a rational preference relation $\succsim$ over the set of all possible alternatives X
 2. the set of possible choice problems D is determined
 3. a choice problem $A \in D$ realizes
 4. the agent's choice is from the set of her most desirable alternative(s) in A , that is, one of the maximal elements in A given $\succsim$, a $\succsim$ -best/optimal choice from A
- ▶ Note that the
 - ▶ Rational economic agent's preferences are shaped before D and A are recognized. $\succsim$ over X is taken as independent of D and A .
 - ▶ Rationality in economics does not involve making any judgments about preferences. A rational agent can have preferences that are universally viewed as being against his best interest.
- ▶ Idea of rational choice reflected in the following definitions...

Choice rule derived from a given preference relation

Definition 6

Given a preference relation $\succsim$, the *induced choice rule* $C_{\succsim}$ is defined by

$$C_{\succsim}(A) = \{x \in A \mid x \succsim y \text{ for all } y \in A\}.$$

- ▶ given $C_{\succsim}$, the decision maker will choose an object that he weakly prefers to all other objects in A
- ▶ in principle, we could have $C_{\succsim}(A) = \emptyset$ for some A
- ▶ however, if $\succsim$ is rational and X is finite, then $C_{\succsim}(A)$ is never empty
- ▶ this follows directly from the fact that if $\succsim$ is rational, then every finite $A \subseteq X$ has a maximal element (see Lemma 1), and a maximal element is in $C_{\succsim}(A)$

Rationalizability of a Choice Rule

Definition 7

A choice rule C is **rationalizable** if there is a rational preference relation $\succsim$ on X such that $C = C_{\succsim}$ (i.e., $C(A) = C_{\succsim}(A)$ for any A in the domain of C).

A preference relation $\succsim$ **rationalizes** C if $C = C_{\succsim}$ and $\succsim$ is rational (i.e., $C(A) = C_{\succsim}(A)$ for any A in the domain of C).

- ▶ a choice rule is rationalizable if she maximizes some rational preference relation among the available alternatives
- ▶ there may be more than one preference relation which rationalizes a given choice rule

Preferences derived from choices

Definition 8

Given a choice rule C , its *revealed preference relation* $\succsim_C$ is defined by

$x \succsim_C y$ if and only if $x, y \in A$ and $x \in C(A)$ for some $A \in D$.

- ▶ we read $x \succsim_C y$ as "x is revealed at least as good as y", if in addition $y \notin C(A)$, we say that "x is revealed strictly better than y"
- ▶ x is revealed at least as good as y iff x is an acceptable choice when y is available, and if in addition, y is not an acceptable choice, then x is revealed strictly better than y
- ▶ note that $\succsim_C$ is not necessarily rational, in particular for $\succsim_C$ to be complete, we must have that for any pair $x, y \in X$,
 $x, y \in A$ and $x \in C(A)$ or $y \in C(A)$ for some $A \in D$

A condition for C to be rationalizable: The Weak Axiom

Definition 9

The choice rule C satisfies the *weak axiom of revealed preference* if the following property holds:

If $x, y \in A \cap B$ and $x \in C(A)$ and $y \in C(B)$, then $x \in C(B)$.

- ▶ In words: if x is an acceptable choice when y is available, then there can be no A containing both x and y for which y is an acceptable choice and x is not
- ▶ An alternative formulation of the weak axiom through the revealed preference relation:
If x is revealed at least as good as y in choice problem A , then y cannot be revealed strictly better than x in choice problem B .

The Weak Axiom implies Sen's α

Definition 10 (Sen's α .)

*A choice rule C satisfies Sen's α if the following property holds:
If $x \in A \subseteq B$ and $x \in C(B)$, then $x \in C(A)$.*

- ▶ "Sen's α " is also referred to as "Independence of Irrelevant Alternatives".
- ▶ Example: If I choose Italy out of all worldwide holiday spots ($I \in C(\text{World})$), then I also choose Italy out of all European holiday spots ($I \in C(\text{Europe})$).
- ▶ If a choice rule C satisfies the WA, then it satisfies Sen's α :
If $x \in C(B)$, then x is revealed at least as good as every $y \in A \subseteq B$. Thus, by the WA, no $y \in A$ can be revealed strictly better than x and since $C(A)$ is non-empty, x must be in $C(A)$.

The Weak Axiom implies Sen's β

Definition 11 (Sen's β .)

A choice rule C satisfies Sen's β if the following property holds:

If $x, y \in A \subseteq B$, $x, y \in C(A)$ and $y \in C(B)$, then $x \in C(B)$.

- ▶ "Sen's β " is sometimes referred to as "Expansion Consistency".
- ▶ Example: If France and Italy are the best European holiday spots for me ($F, I \in C(Europe)$) and Italy is one of the best worldwide holiday spots for me ($I \in C(World)$), then France must also be a best worldwide holiday spot for me ($F \in C(World)$).
- ▶ If a choice rule C satisfies the WA, then it satisfies Sen's β :
Since $x, y \in C(A)$, x is revealed at least as good as y . By the WA, y cannot be revealed strictly better than x . Thus x must be in $C(B)$ as well.